JONGIN CHOI

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EDUCATION

Pennsylvania State University ***May 2025***
Bachelor of Science, Applied Data Sciences *University Park, PA*

- Coursework: Data Analysis, Software Engineering, Statistics, Algorithms, Artificial Intelligence, Machine Learning

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, TypeScript
Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, HuggingFace Transformers, Jupyter, Spark
Web & Scraping: Playwright, Beautiful Soup, HTML/CSS
Databases & Tools: MySQL, MongoDB, Git, Linux, Docker

EXPERIENCE

Research Assistant — Multimodal Emotion Dataset (FeelReel) ***Jun 2024 – Mar 2025***
Pennsylvania State University, Data Science Lab (Prof. Sharon Hwang) · IST 495 internship credit *University Park, PA*

- Supported the follow-up phase of FeelReel, a lab benchmark of 400K+ Reddit images and 5M+ comments for emotion understanding (earlier version submitted to CVPR 2024).
- Sourced and organized Reddit image posts and comments through the project's data collection pipeline, keeping datasets structured and reproducible.
- Built the MTurk survey (HIT) page and collected human evaluations of the dataset's emotion annotations.

PROJECTS

FeelReel-Mini — Personal recreation ***Jul – Oct 2026***
github.com/nopgae/feelreel-mini *PyTorch, Hugging Face, PRAW, LLaVA (Ollama), matplotlib*

- Rebuilt the FeelReel data pipeline at small scale with public tools: Reddit image posts and comments, GoEmotions pseudo-labels (RoBERTa, 28 emotions), and LLaVA 7B asked what in each image triggered the emotion.
- Charted emotion distributions across 8 image-heavy subreddits: scenery and woodworking communities leaned toward admiration and surprise, pet communities toward joy.
- Separate from the lab's code and data, which are not public.

Medical Transcription Classifier — Penn State DS 310 ***Fall 2024***

- Classified medical transcriptions into 6 specialties with TF-IDF (1–3 grams) and document-length features; weighted F1 0.807.
- Tuned XGBoost over 54 configurations with 5-fold CV (270 fits), then showed untuned logistic regression still won (F1 0.763 vs 0.711) and traced it to sparse features.

Reddit Sentiment Pipeline — Personal ***Aug 2025 – Mar 2026***
github.com/nopgae/reddit-sentiment-roberta *PyTorch, Hugging Face, pandas, Plotly*

- Built an end-to-end pipeline alone: Reddit JSON API collection, batched inference with a pre-trained RoBERTa sentiment model, aggregation and charts.
- Scored 563 posts across 7 e-commerce, logistics and supply-chain subreddits; quantified community baselines (r/supplychain +0.11 vs r/ecommerce –0.31).

Starbucks Korea Parking Finder — Personal ***Apr 2026 (tests/CI Oct 2026)***
github.com/nopgae/starbucks-parking-finder *TypeScript, Playwright, Fastify, PostgreSQL, GitHub Actions*

- Used Playwright to find the JSON request behind the store locator, then called it directly to collect parking info for all 2,131 stores (1,514 with parking details) instead of checking each store page by hand.
- Wrote rule-based parsers that turn free-text parking notes into free / paid / free-with-purchase / none plus capacity and location, and turn queries like "free parking 2+ hours near Gangnam Station" into filters; served via a Fastify API and map UI, with parser unit tests run on every push by GitHub Actions.

ADDITIONAL INFORMATION

Languages: Fluent in English and Korean (bilingual); TOEIC Speaking 160 (AL)
Interests: Web security CTF (webhacking.kr), Boston Dynamics Spot robotics